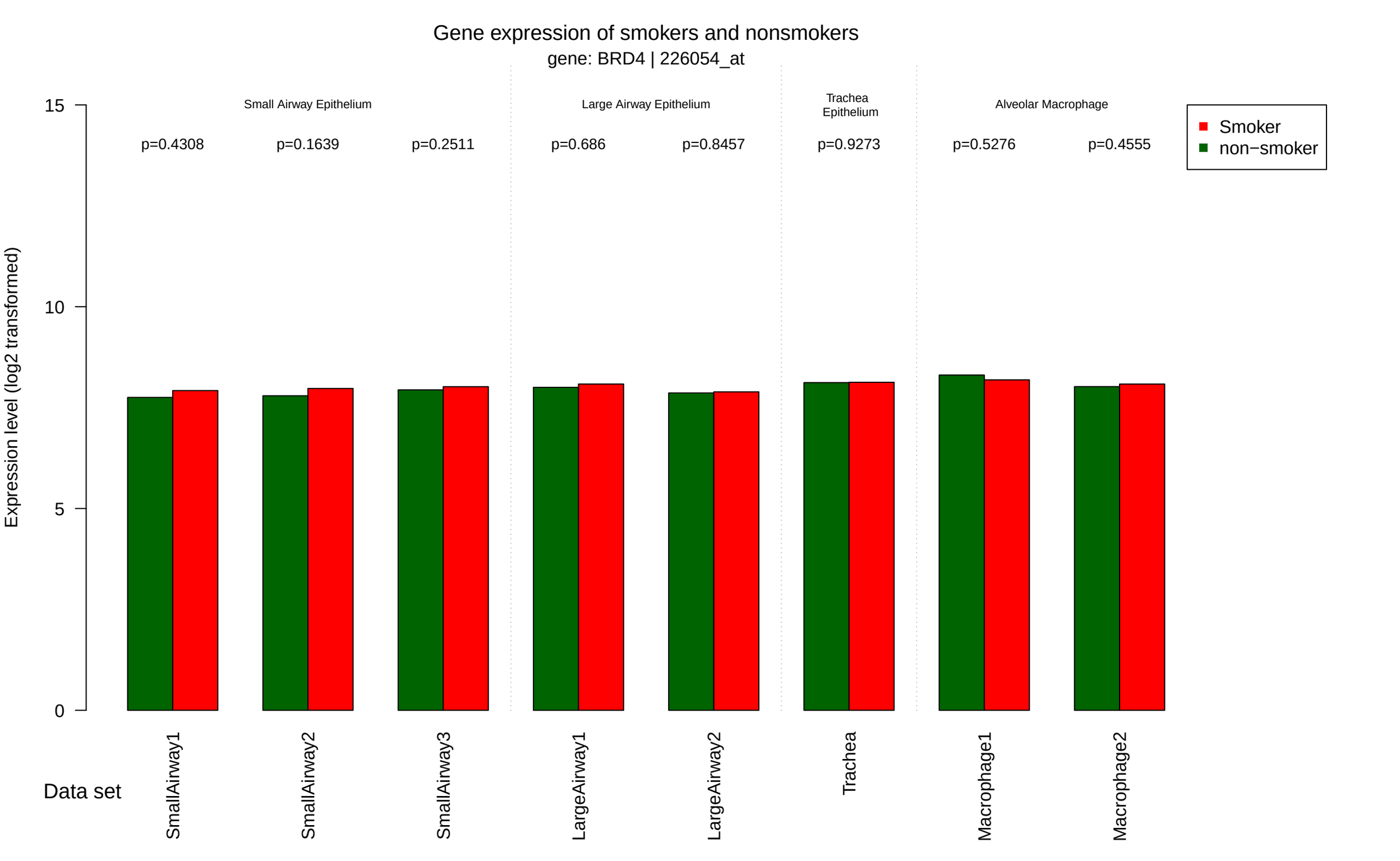
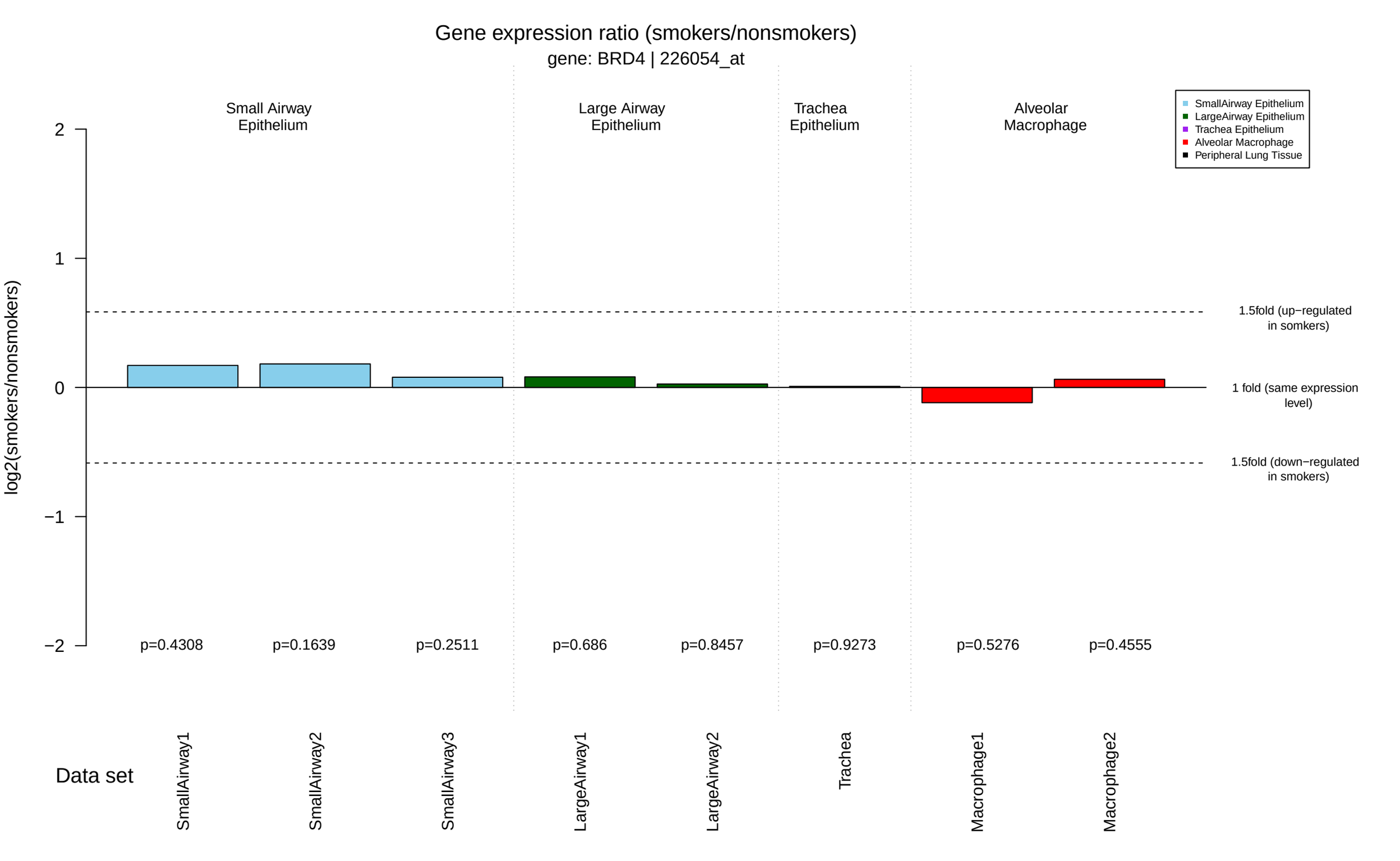
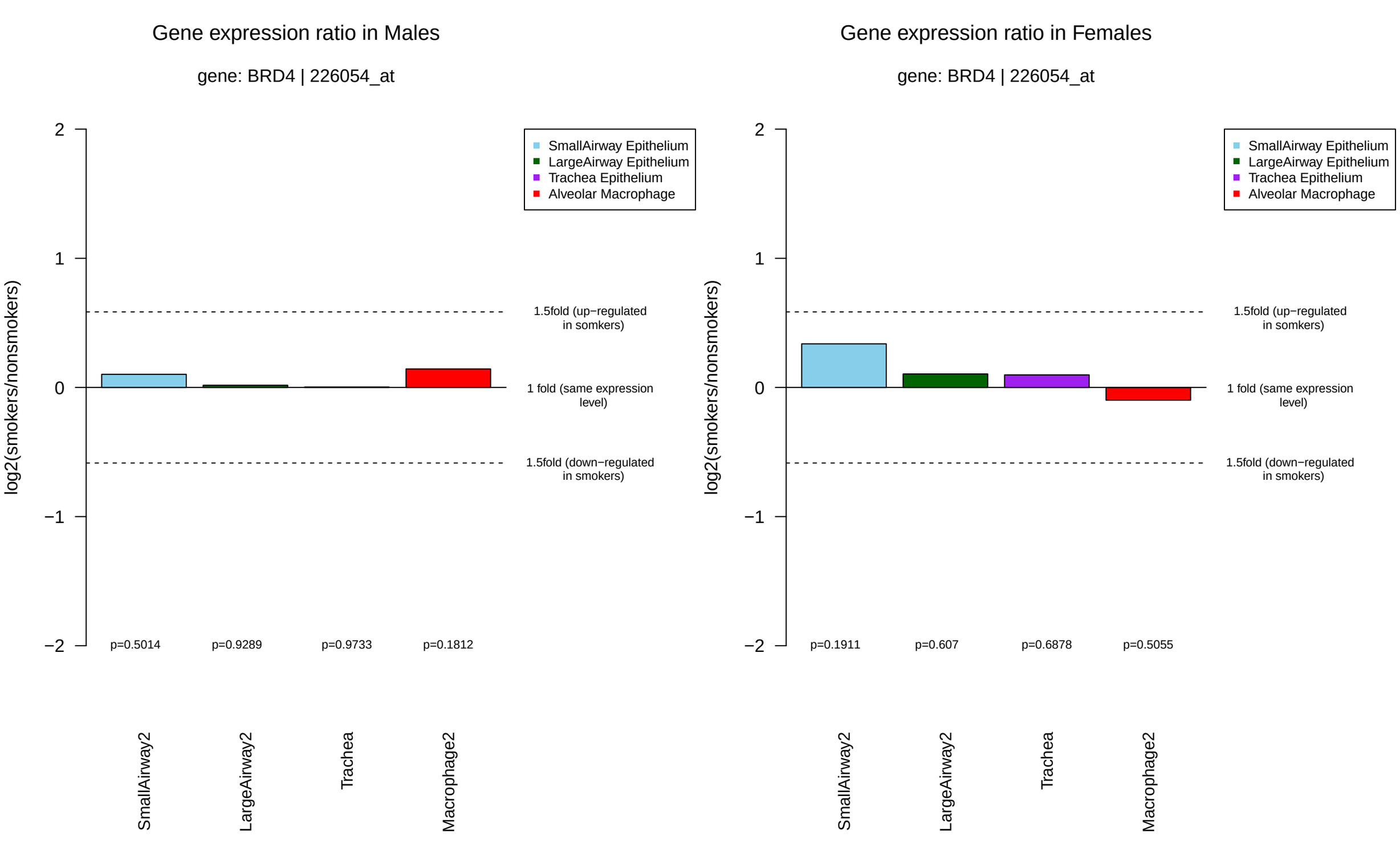
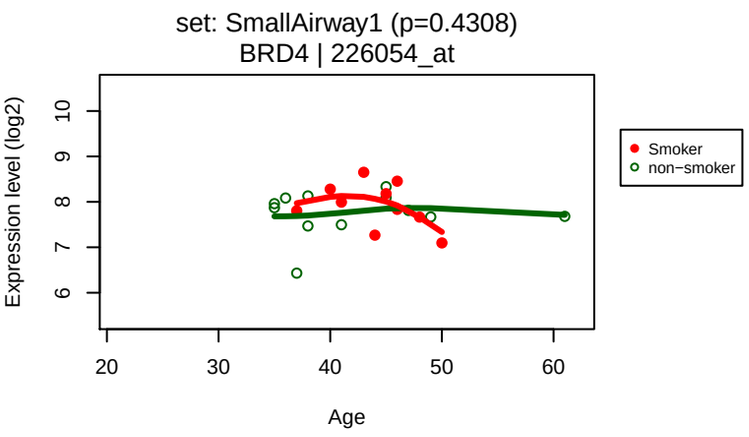
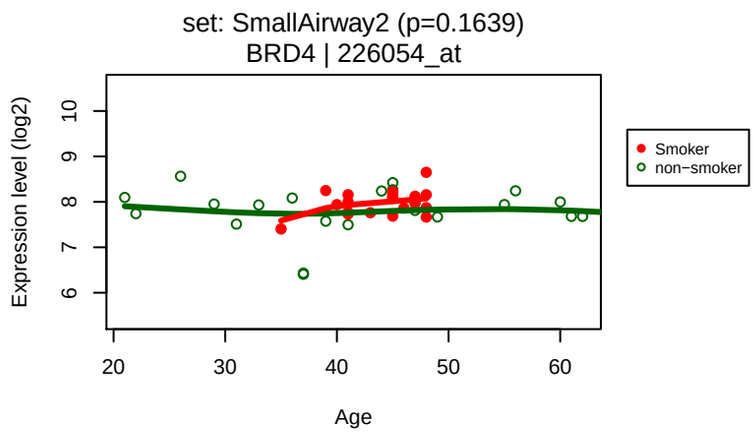
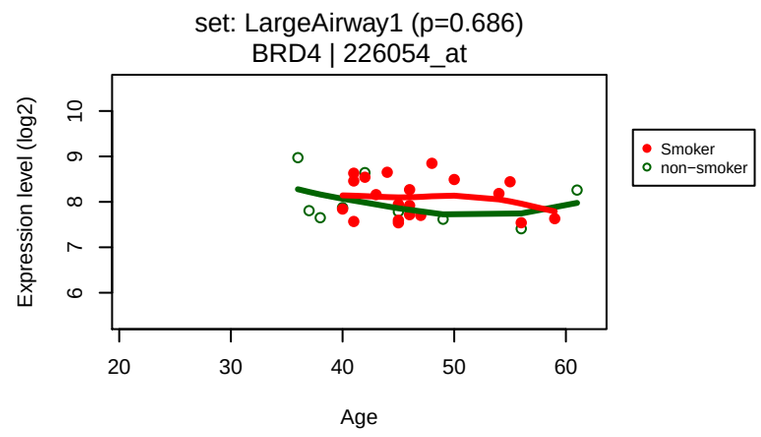

Gene expression of smokers and nonsmokers
gene: BRD4 | 226054_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4308
p=0.1639
p=0.2511
p=0.686
p=0.8457
p=0.9273
p=0.5276
p=0.4555
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

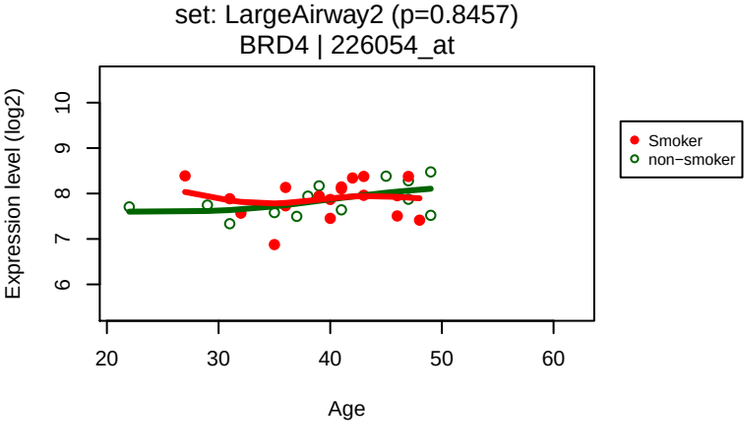

Gene expression ratio (smokers/nonsmokers)
gene: BRD4 | 226054_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4308
p=0.1639
p=0.2511
p=0.686
p=0.8457
p=0.9273
p=0.5276
p=0.4555
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

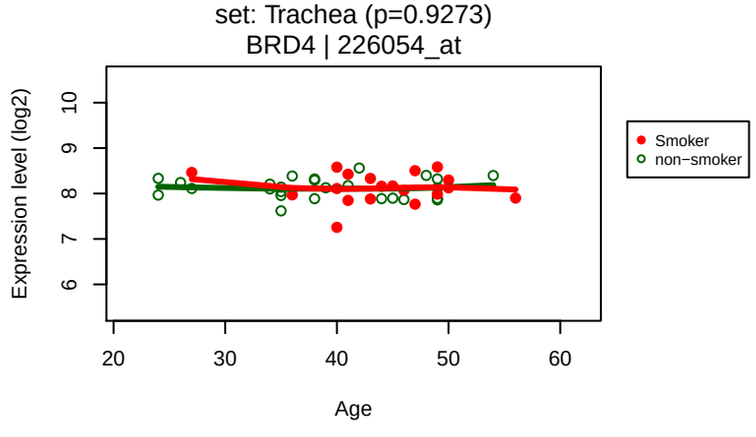

Gene expression ratio in Males
gene: BRD4 | 226054_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5014
p=0.9289
p=0.9733
p=0.1812
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRD4 | 226054_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1911
p=0.607
p=0.6878
p=0.5055
SmallAirway2
LargeAirway2
Trachea
Macrophage2

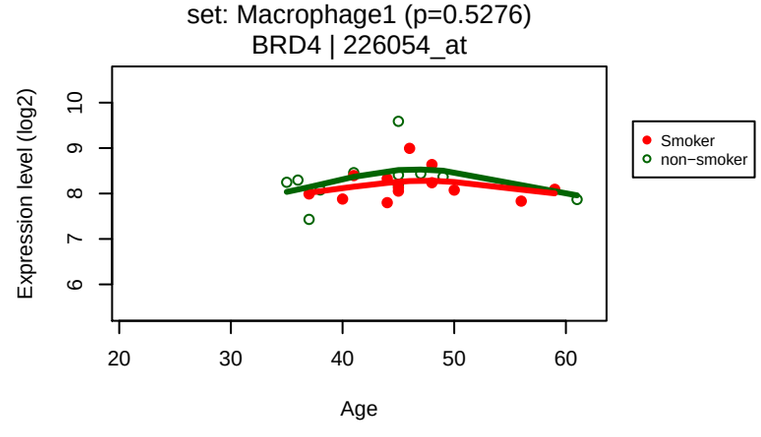

set: SmallAirway1 (p=0.4308)
BRD4 | 226054_at
Expression level (log2)
Age
Smoker
non−smoker

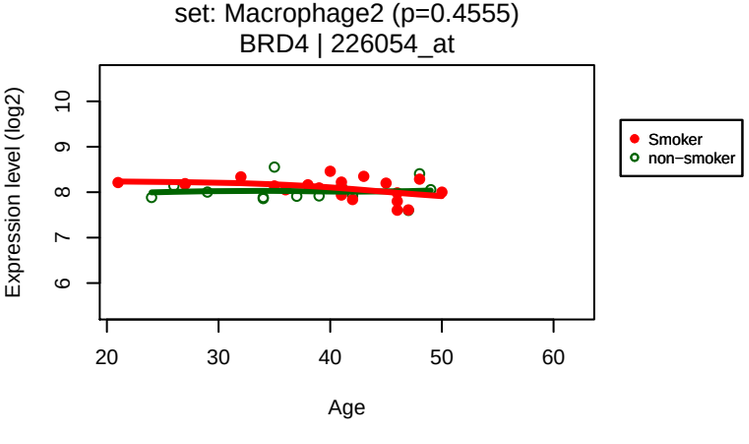

set: SmallAirway2 (p=0.1639)
BRD4 | 226054_at
Expression level (log2)
Age
Smoker
non−smoker

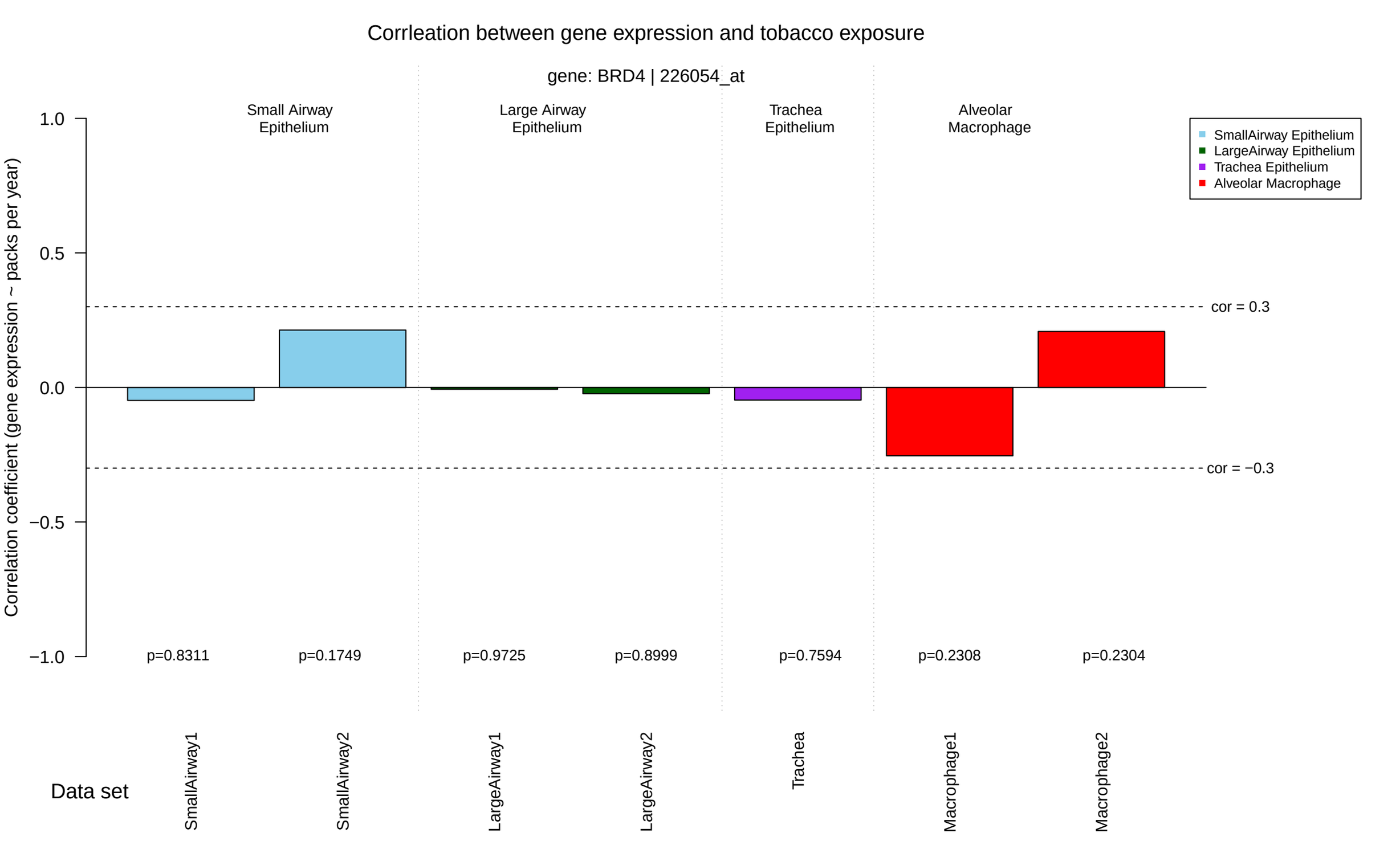

set: LargeAirway1 (p=0.686)
BRD4 | 226054_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8457)
BRD4 | 226054_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9273)
BRD4 | 226054_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5276)
BRD4 | 226054_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4555)
BRD4 | 226054_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRD4 | 226054_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8311
p=0.1749
p=0.9725
p=0.8999
p=0.7594
p=0.2308
p=0.2304
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2